

6.9 Missing transverse momentum

The incoming partons at the LHC carry little transverse momentum, so the vector sum of the transverse momenta of all the particles is expected to be approximately zero. An imbalance in the visible transverse momentum can therefore indicate undetected particles, such as neutrinos or weakly interacting BSM particles. In this analysis, the neutrino produced in the LQ process and the tau-decay neutrino can together give rise to sizeable missing transverse momentum ($\vec{p}_T^{\text{miss}}$). $\vec{p}_T^{\text{miss}}$ is reconstructed as the negative vector sum of the calibrated visible contributions, with its magnitude denoted by E_T^{miss} [1]:

$$\vec{p}_T^{\text{miss}} = - \left(\sum_e \vec{p}_T^e + \sum_\mu \vec{p}_T^\mu + \sum_{\tau_{\text{had}}} \vec{p}_T^\tau + \sum_j \vec{p}_T^j + \vec{p}_T^{\text{soft}} \right), \quad (6.3)$$

$$E_T^{\text{miss}} = \left| \vec{p}_T^{\text{miss}} \right|. \quad (6.4)$$

Two types of contribution are included in the reconstruction:

- **hard term:** The electron, muon, hadronic-tau and jet contributions selected for $\vec{p}_T^{\text{miss}}$ reconstruction. The jet contribution is selected using the Tight $\vec{p}_T^{\text{miss}}$ working point to suppress pile-up jets. This requires jet $p_T > 20$ GeV for $|\eta| < 2.5$ and $p_T > 30$ GeV for $2.5 \leq |\eta| < 4.5$.
- **track soft term:** The vector sum of transverse momenta of selected PV tracks not used in the hard term.

6.10 Object definitions in the analysis

Baseline and signal objects are defined to provide two levels of selection in the analysis. After overlap removal as described in Section 6.8, baseline objects are used for object counting and lepton vetoes. Signal objects are selected from these baseline collections with tighter identification or isolation requirements, as summarised in Table 6.1.

Table 6.1 Baseline object definitions and additional signal requirements used in the analysis. A dash in the signal column indicates that the requirement depends on the analysis region.

Object	Requirement	Baseline	Additional signal requirement
Electron	p_T	$> 10 \text{ GeV}$	—
	$ \eta $	$(\eta < 1.37) \vee (1.52 < \eta < 2.47)$	
	$ d_0 /\sigma_{d_0}$		< 5
	$ \Delta z_0 \sin \theta $		$< 0.5 \text{ mm}$
	Identification		TightLH
	Isolation	Loose_VarRad	HighPtCaloOnly
Muon	p_T	$> 10 \text{ GeV}$	—
	$ \eta $		< 2.5
	$ d_0 /\sigma_{d_0}$		< 3
	$ \Delta z_0 \sin \theta $		$< 0.5 \text{ mm}$
	Identification	Loose	Medium
	Isolation	Loose_VarRad	Tight_VarRad
Hadronic tau	$ \eta $	$(\eta < 1.37) \vee (1.52 < \eta < 2.5)$	
	Tracks		1, 3
	Charge		$ Q = 1$
	Energy calibration		MVA
	p_T	$> 20 \text{ GeV}$	—
	Jet rejection	RNN, Loose	RNN, Tight
	Electron rejection		RNN, Loose
Jet	Algorithm		Anti- k_t
	Radius		$R = 0.4$
	Inputs		PFlow
	Cleaning		LooseBad
	p_T	$> 20 \text{ GeV}$	—
	$ \eta $	< 2.5	—
	NNJVT		FixedEffPt
	b -tagging	GN2v01, 85% b -jet efficiency	

References

- 24 [1] ATLAS Collaboration. The performance of missing transverse momentum reconstruction
25 and its significance with the ATLAS detector using 140 fb^{-1} of $\sqrt{s} = 13 \text{ TeV}$ pp collisions.
26 *Eur. Phys. J. C*, 85:606, 2025. doi: 10.1140/epjc/s10052-025-14062-8.